1 Summary

Model (Run-name): model1 (test)

Iteration: 29

TAC: 154883.75523012146

Took: 2.0815224647521973s

Total Time: 58.6903178691864s

Epsilons:

- ε_{LMTD}: 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area}: 1e-06
- ε_{Beta}: 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: `http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html`

2 Results

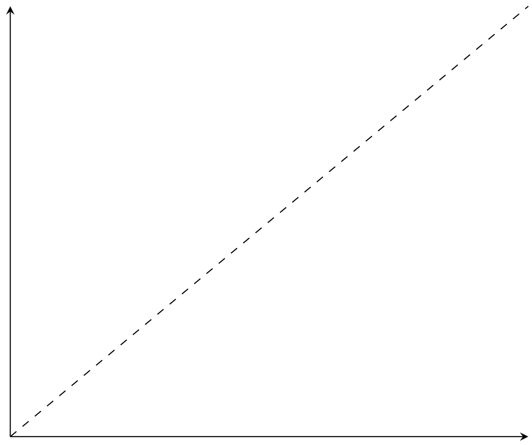
Variable	Value
$z_{1,1,1}$	1.0
$z_{1,2,2}$	1.0
$z_{2,1,2}$	1.0
$z_{cu,1}$	1.0
$z_{cu,2}$	1.0
$z_{hu,1}$	1.0
$A^{\beta}_{1,1,1}$	70.58322282660124
$A^{\beta}_{1,2,2}$	68.20353230192663
$A^{\beta}_{2,1,2}$	144.42095395765313
$A^{\beta}_{cu,1}$	5.0892956597199035
$A^{\beta}_{cu,2}$	37.63325504358228
$A^{\beta}_{hu,1}$	13.195607617969486
$A_{1,1,1}$	70.58322282660124
$A_{1,2,2}$	68.20353230192663
$A_{2,1,2}$	144.42095395765313
$A_{cu,1}$	5.0892956597199035
$A_{cu,2}$	37.63325504358228
$A_{hu,1}$	13.195607617969486
$dt_{1,1,1}$	32.466579030879686
$dt_{1,1,2}$	10.0
$dt_{1,2,2}$	82.60026290736096
$dt_{1,2,3}$	37.60026290736093
$dt_{2,1,2}$	17.399737092639043
$dt_{2,1,3}$	58.04980281947928
$dt_{cu,1}$	67.60026290736093
$dt_{cu,2}$	148.04980281947928
$dt_{hu,1}$	62.466579030879686
$f^{H}_{1,1,1}$	2.407133467594252
$f^{H}_{1,2,2}$	9.740535833942998
$f^{H}_{2,1,2}$	11.08638156186552
$f^{C}_{1,1,1}$	11.45502905162682
$f^{C}_{1,2,2}$	13.0
$f^{C}_{2,1,2}$	10.497357145298306

Variable	Value
$RecLMTD_{1,1,1}$	0.052416783444583025
$RecLMTD_{1,2,2}$	0.017488085205622228
$RecLMTD_{2,1,2}$	0.029639268631382065
$RecLMTD_{cu,1}$	0.014536323589406201
$RecLMTD_{cu,2}$	0.009597121429486133
$RecLMTD_{hu,1}$	0.02259043820592644
$q_{1,1,1}$	673.9973709263907
$q_{1,2,2}$	1950.0
$q_{2,1,2}$	2439.0039436104144
$q_{cu,1}$	176.00262907360934
$q_{cu,2}$	1960.9960563895856
$q_{hu,1}$	486.99868546319533
$t^{(H)}_{1,1}$	650.0
$t^{(H)}_{1,2}$	582.600262907361
$t^{(H)}_{1,3}$	387.60026290736096
$t^{(H)}_{2,1}$	590.0
$t^{(H)}_{2,2}$	590.0
$t^{(H)}_{2,3}$	468.0498028194793
$t^{(C)}_{1,1}$	617.5334209691204
$t^{(C)}_{1,2}$	572.600262907361
$t^{(C)}_{1,3}$	410.0
$t^{(C)}_{2,1}$	500.0
$t^{(C)}_{2,2}$	500.0
$t^{(C)}_{2,3}$	350.0
$t^{H}_{1,1,1}$	370.0
$t^{H}_{1,2,2}$	382.4080179515642
$t^{H}_{2,1,2}$	370.0
$t^{C}_{1,1,1}$	628.394494805876
$t^{C}_{1,2,2}$	500.0
$t^{C}_{2,1,2}$	642.3445711002439
$b^{H,\mathrm{in}}_{1,1,1}$	1564.6367539362639
$b^{H,\mathrm{in}}_{1,2,2}$	5674.859002044328
$b^{H,\mathrm{in}}_{2,1,2}$	6540.965121500657

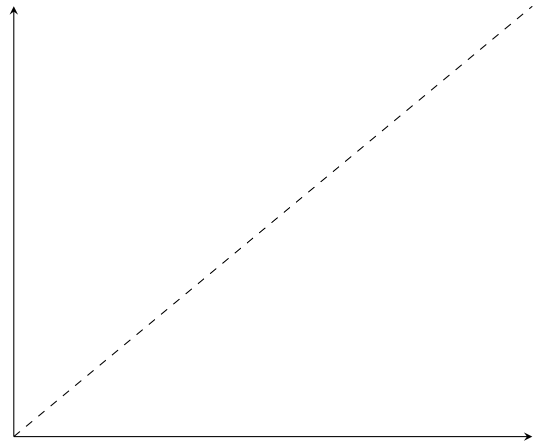
Variable	Value
$b^{H,\mathrm{out}}_{1,1,1}$	890.6393830098733
$b^{H,\mathrm{out}}_{1,2,2}$	3724.8590020443285
$b^{H,\mathrm{out}}_{2,1,2}$	4101.961177890243
$b^{C,\mathrm{in}}_{1,1,1}$	6558.879608643174
$b^{C,\mathrm{in}}_{1,2,2}$	4550.0
$b^{C,\mathrm{in}}_{2,1,2}$	4303.916429572306
$b^{C,\mathrm{out}}_{1,1,1}$	7232.876979569564
$b^{C,\mathrm{out}}_{1,2,2}$	6500.0
$b^{C,\mathrm{out}}_{2,1,2}$	6742.920373182719

3 *RecLMTD* Tangent Points

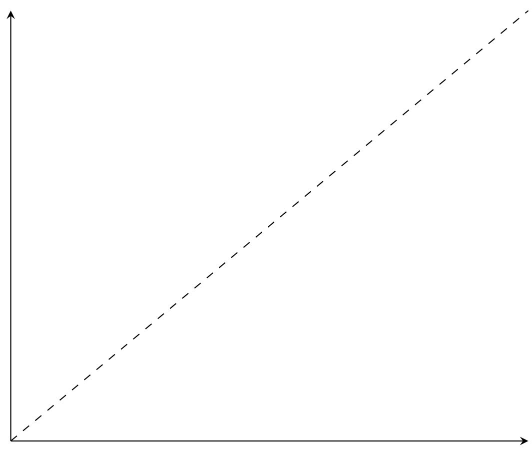
- $(1,1,1)$
  - Value: 0.052417
  - Error: 0.000000014329117
  - Relative error: 0.000000273368800
- $(1,2,2)$
  - Value: 0.017488
  - Error: 0.000000844146183
  - Relative error: 0.000048267459164
- $(2,1,2)$
  - Value: 0.029639
  - Error: 0.000000196726697
  - Relative error: 0.000006637322724
- $cu, 1$
  - Value: 0.014536
  - Error: 0.000000006224747
  - Relative error: 0.000000428220012
- $cu, 2$
  - Value: 0.009597
  - Error: 0.000000000582199
  - Relative error: 0.000000060663888
- $hu, 1$
  - Value: 0.022590
  - Error: 0.000000001515323
  - Relative error: 0.000000067078049



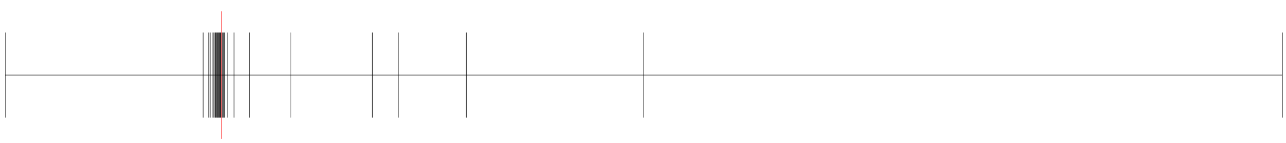
(a) *RecLMTD* - $(1, 1, 1)$



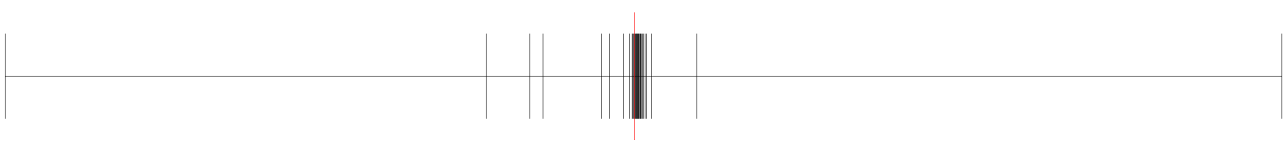
(b) *RecLMTD* - $(1, 2, 2)$



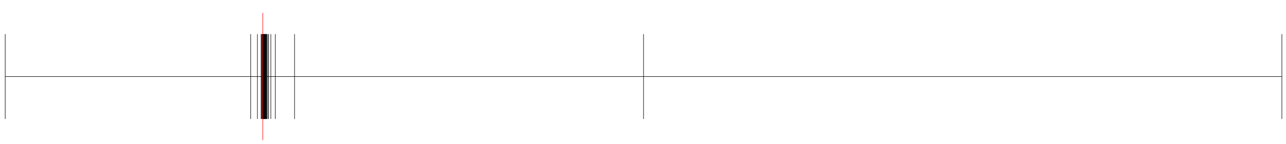
(c) *RecLMTD* - $(2, 1, 2)$



(a) *RecLMTD* - $cu, 1$



(b) *RecLMTD* - $cu, 2$



(a) *RecLMTD* - $hu, 1$

4 Balancing Breakpoints

4.1 $b^{\boldsymbol{H},\mathbf{in}}$

- $(1, 1, 1)$
  - Value: 1564.6367539362639
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 5674.859002044328
  - Absolute error: 0.020264330567442812
  - Relative error: 3.5709086203226216e-06
- $(2, 1, 2)$
  - Value: 6540.965121500657
  - Absolute error: 0.0
  - Relative error: 0.0

4.2 $b^{\boldsymbol{H},\mathbf{out}}$

- $(1, 1, 1)$
  - Value: 890.6393830098733
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 3724.8590020443285
  - Absolute error: 0.0
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 4101.961177890243
  - Absolute error: 0.0
  - Relative error: 0.0

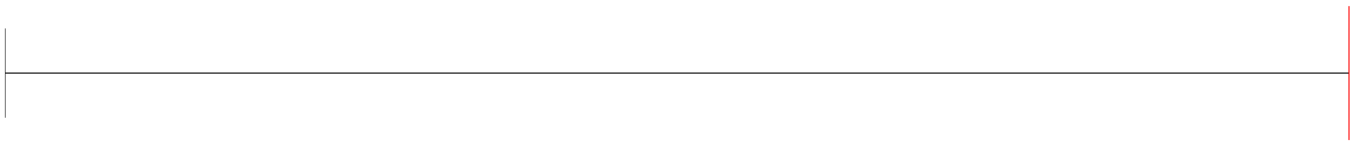
4.3 $b^{\boldsymbol{C},\mathbf{in}}$

- $(1, 1, 1)$
  - Value: 6558.879608643174
  - Absolute error: 0.2730379298009211
  - Relative error: -4.1627012590350054e-05
- $(1, 2, 2)$
  - Value: 4550.0
  - Absolute error: 0.0

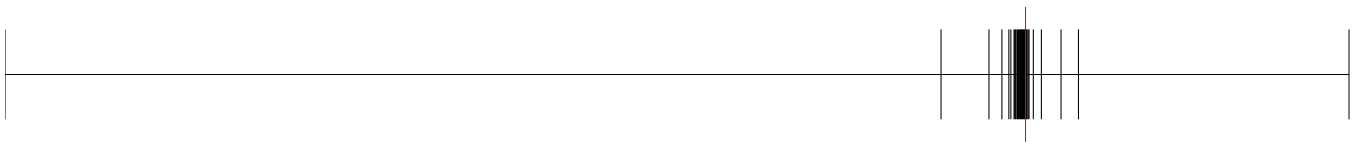
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 4303.916429572306
  - Absolute error: 0.0
  - Relative error: 0.0

4.4 $b^{\mathbf{C,out}}$

- $(1, 1, 1)$
  - Value: 7232.876979569564
  - Absolute error: 34.59978568589668
  - Relative error: 0.004806675924524816
- $(1, 2, 2)$
  - Value: 6500.0
  - Absolute error: 0.0
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 6742.920373182719
  - Absolute error: 1.8189894035458565e-12
  - Relative error: -2.6976284797610275e-16

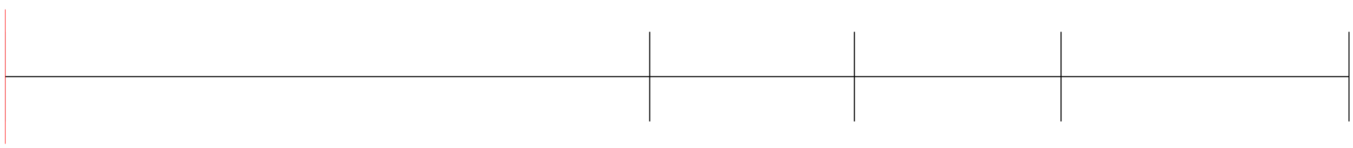


(a) $t^{(H)}$ - $(1,1)$



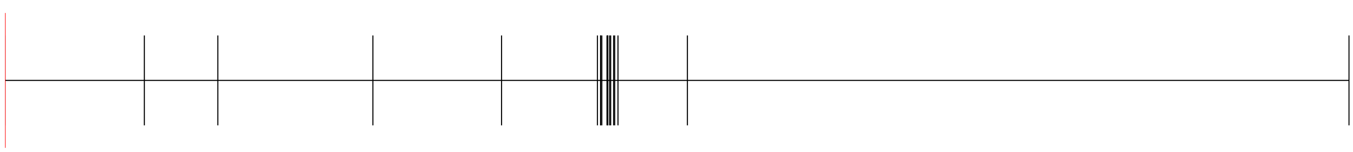
(a) $t^{(H)}$ - $(1,2)$

(a) $t^{(H)}$ - $(2,2)$



(a) $t^{(H)}$ - $(1,1,1)$

(a) $t^{(H)}$ - $(1,2,2)$



(a) $t^{(H)}$ - $(2,1,2)$

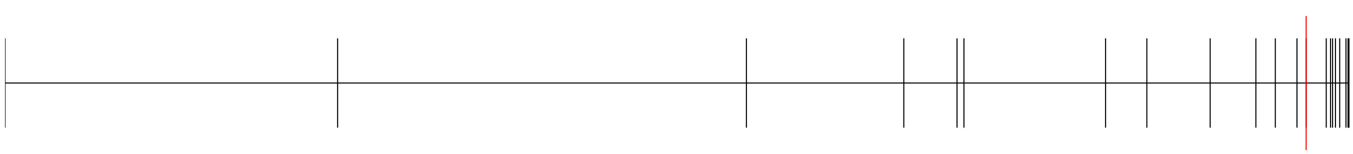
(a) $t^{(C)}$ - $(1,2)$

(a) $t^{(C)}$ - $(2,3)$

(a) $t^{(C)}$ - $(1,3)$

(a) $t^{(C)}$ - $(1,1,1)$

(a) $t^{(C)}$ - $(1,2,2)$



(a) $t^{(C)}$ - $(2,1,2)$

5 Area Breakpoints

5.1 A

- $(1, 1, 1)$
  - Value: 70.583223
  - Error: 0.074325641532596
  - Relative error: 0.001051913675807
- $(1, 2, 2)$
  - Value: 68.203532
  - Error: 0.000000000000057
  - Relative error: 0.000000000000001
- $(2, 1, 2)$
  - Value: 144.420954
  - Error: 0.159632197685482
  - Relative error: 0.001104105343120

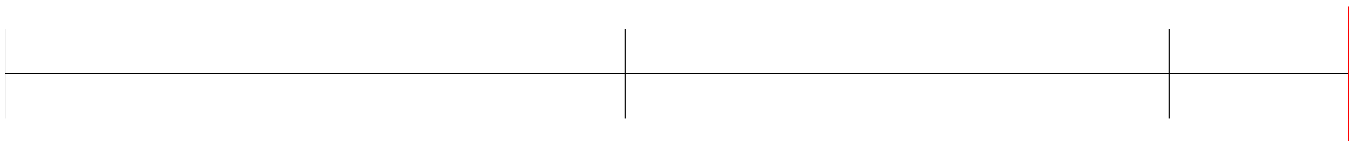
5.2 A_{CU}

- $cu, 1$
  - Value: 5.089296
  - Error: 0.027566677880531
  - Relative error: 0.005387418316487
- $cu, 2$
  - Value: 37.633255
  - Error: 0.006579508246297
  - Relative error: 0.000174801731321

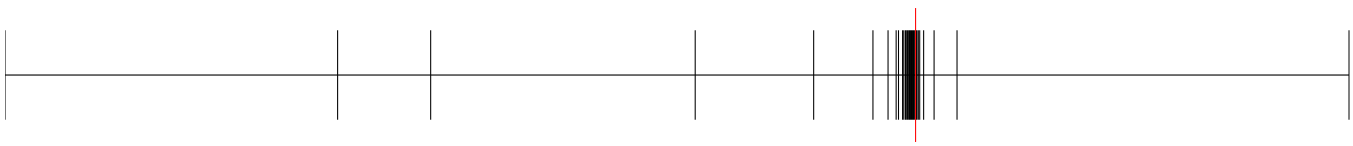
5.3 A_{HU}

- $hu, 1$
  - Value: 13.195608
  - Error: 0.006208834418979
  - Relative error: 0.000470301525655

(a) A^{β} - $(1,1,1)$

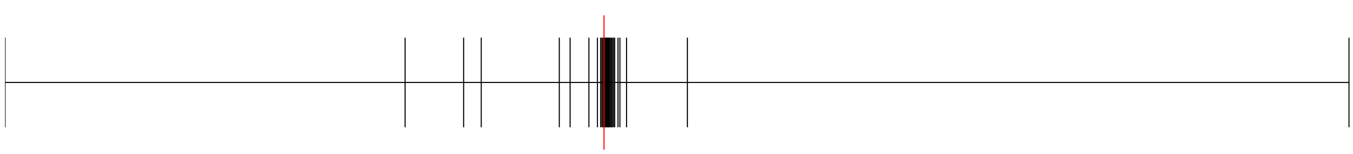


(a) A^{β} - $(1,2,2)$



(a) A^{β} - $(2,1,2)$

(a) A^{β} - $cu, 1$



(a) A^{β} - $cu, 2$

(a) A^{β} - $hu, 1$

6 A^{β} Breakpoints

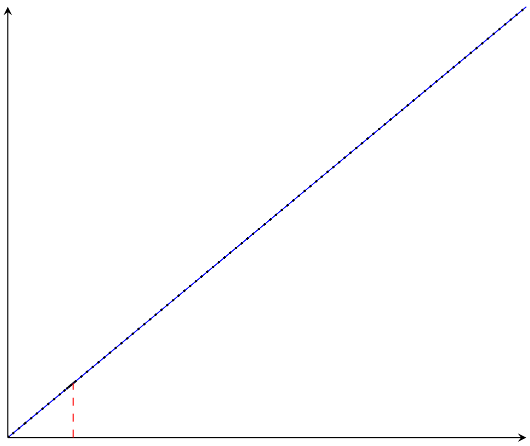
- $(1,1,1)$
  - q: 70.583223
  - $\hat{q}^{\beta}$: 70.583223
  - q^{β}: 70.583223
  - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.11420238741596, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.60541450524272, 70.65581083979713, 70.83181816751056, 70.95870595008942, 71.00935653192329, 71.18527282143235, 71.22933662920877, 71.2616536557244, 71.33750386361461, 71.62767096549581, 71.6637831223933, 71.88123217857378, 72.01156079815767, 72.36161285231599, 560.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $(1,2,2)$
  - q: 68.203532
  - $\hat{q}^{\beta}$: 68.203532
  - q^{β}: 68.203532
  - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, 67.60071888621576, 67.91344539650363, 68.10969007738701, 68.48329505760876, 68.66676327863097, 68.85281320938876, 69.0314457108091, 69.22655548799992, 69.34502840170047, 69.41355730879985, 69.60484346150977, 70.08419437796795, 70.37078617087542, 70.56626459266522, 70.94081287188642, 71.26369181826138, 72.25348351595485, 72.87506706799363, 74.6496383881616, 84.1468297226002, 390.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $(2,1,2)$
  - q: 144.420954
  - $\hat{q}^{\beta}$: 144.420954
  - q^{β}: 144.420954
  - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 127.07690270456857, 128.52021081477858, 131.69760457629607, 132.12228195956155, 132.5928424015674, 135.25996987589937, 135.81963163113545, 137.02923097052596, 138.86359281280824, 139.47623233772285, 140.31685261310218, 140.7008934199032, 141.79783607898793, 142.4338700260237, 143.30274961189664, 144.67005099128556, 144.70843093515643, 147.02575214945074, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $cu, 1$
  - q: 5.089296
  - $\hat{q}^{\beta}$: 5.089296
  - q^{β}: 5.089296
  - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.0691182452509045, 4.269280092716329, 4.432280503220532, 4.440201431981878, 4.600843803756838, 4.613047202566896, 4.707108999269609, 4.813467676447643, 4.825601156120121, 4.849719301830286, 4.888121952381175, 4.962276317638335,

4.974201802143407, 4.983938626310074, 4.993068227518925, 5.0359974771272995, 5.0418124100090695, 5.081618324442152, 5.195781532076625, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]

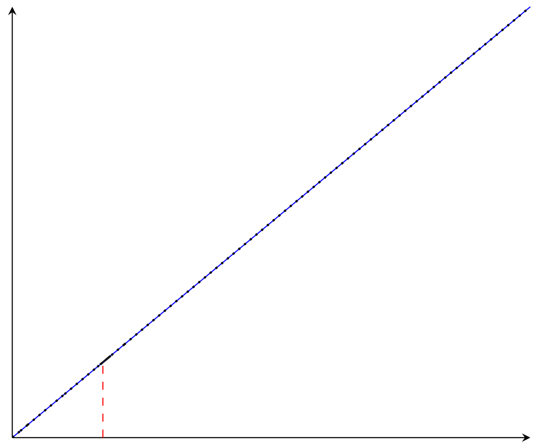
- Error: 0.000000000000000
- Relative error: 0.000000000000000

- $cu, 2$
  - q: 37.633255
  - $\hat{q}^\beta$: 37.633255
  - q^β: 37.633255
  - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, 37.52720514437563, 37.554741031910254, 37.612770977560544, 37.65435035996605, 37.6551299822246, 37.682669717644174, 37.71812813120871, 37.76995088018704, 37.78573239590169, 37.821651654621526, 37.88148462263016, 37.9018730666985, 37.9454585315849, 37.95085342166671, 38.035748486883605, 38.10872593446926, 38.21652191748154, 38.224970554765775, 38.88483572219945, 285.4001551947793]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000

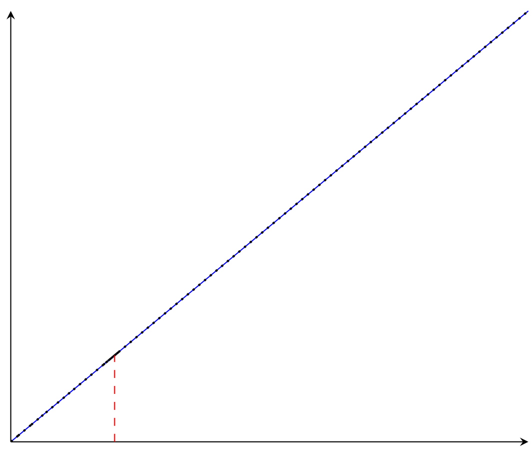
- $hu, 1$
  - q: 13.195608
  - $\hat{q}^\beta$: 13.195608
  - q^β: 13.195608
  - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, 13.134779411775817, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.211215176722128, 13.213749911872785, 13.218222920629291, 13.22717920580809, 13.238585544475612, 13.244985069124171, 13.266498143734903, 13.269097009439808, 13.294574825457861, 13.30871858348944, 13.32330648750218, 13.32663103908035, 13.346967780023082, 13.363827419897227, 13.411355634100326, 13.439500729482463, 13.468285806082731, 13.498843422507207, 14.09178614751222, 237.3002543523117]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000



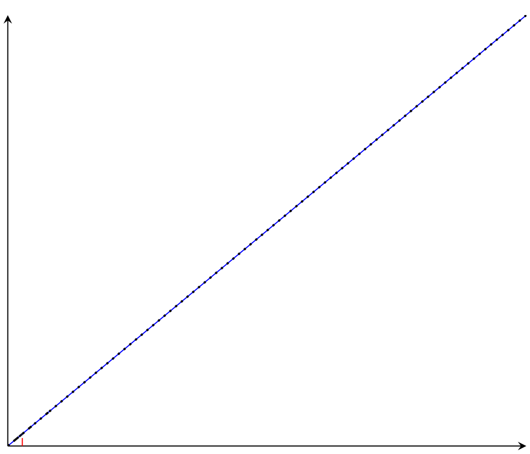
(a) q^β - $(1,1,1)$



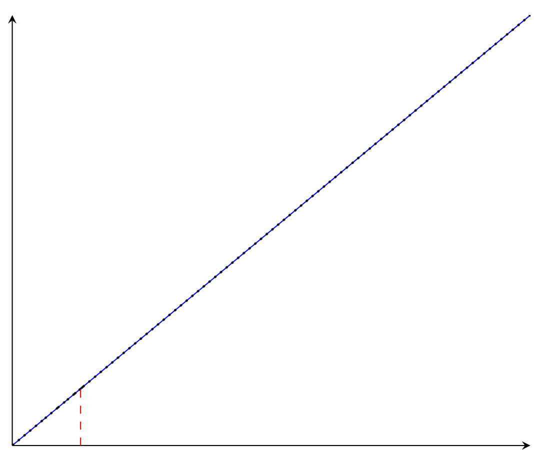
(b) q^β - $(1,2,2)$



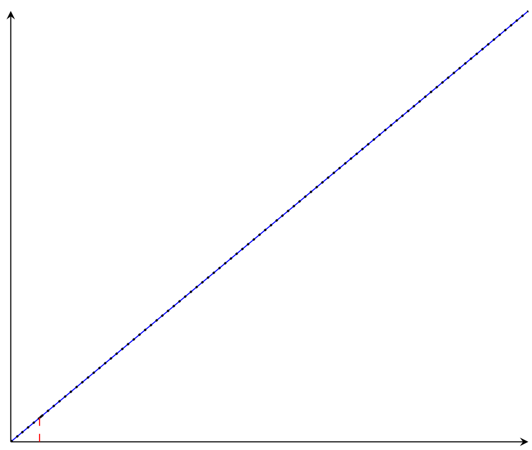
(c) q^β - $(2,1,2)$



(a) q^β - $cu, 1$



(b) q^β - $cu, 2$



(a) q^β - $hu, 1$

7 Other variables

Variable	Value
$z_{1,1,2}$	-0.0
$z_{1,2,1}$	-0.0
$z_{2,1,1}$	0.0
$z_{2,2,1}$	-0.0
$z_{2,2,2}$	-0.0
$z_{hu,2}$	-0.0
$A^{\beta}_{1,1,2}$	0.0
$A^{\beta}_{1,2,1}$	0.0
$A^{\beta}_{2,1,1}$	0.0
$A^{\beta}_{2,2,1}$	0.0
$A^{\beta}_{2,2,2}$	0.0
$A^{\beta}_{hu,2}$	0.0
$A_{1,1,2}$	0.0
$A_{1,2,1}$	0.0
$A_{2,1,1}$	0.0
$A_{2,2,1}$	0.0
$A_{2,2,2}$	0.0
$A_{hu,2}$	0.0
$dt_{1,1,3}$	240.0
$dt_{1,2,1}$	150.0
$dt_{2,1,1}$	16.766929574027625
$dt_{2,2,1}$	90.0
$dt_{2,2,2}$	31.33528252862242
$dt_{2,2,3}$	240.0
$dt_{hu,2}$	180.0
$f^H_{1,1,2}$	0.25946416605700257
$f^H_{1,2,1}$	7.592866532405748
$f^H_{2,1,1}$	20.0
$f^H_{2,2,1}$	0.0
$f^H_{2,2,2}$	8.91361843813448
$f^C_{1,1,2}$	4.502642854701694
$f^C_{1,2,1}$	13.0
$f^C_{2,1,1}$	3.54497094837318

Variable	Value
$f^C_{2,2,1}$	0.0
$f^C_{2,2,2}$	0.0
$RecLMTD_{1,1,2}$	0.1
$RecLMTD_{1,2,1}$	0.008062429492042436
$RecLMTD_{2,1,1}$	0.029166666666666688
$RecLMTD_{2,2,1}$	0.012117270959084082
$RecLMTD_{2,2,2}$	0.007317270959084084
$RecLMTD_{hu,2}$	0.005549329826813255
$q_{1,1,2}$	0.0
$q_{1,2,1}$	0.0
$q_{2,1,1}$	0.0
$q_{2,2,1}$	0.0
$q_{2,2,2}$	0.0
$q_{hu,2}$	0.0
$t^H_{1,1,2}$	565.0836764717336
$t^H_{1,2,1}$	650.0
$t^H_{2,1,1}$	590.0
$t^H_{2,2,1}$	590.0
$t^H_{2,2,2}$	586.5432352842024
$t^C_{1,1,2}$	410.0
$t^C_{1,2,1}$	500.0
$t^C_{2,1,1}$	573.0363330815413
$t^C_{2,2,1}$	350.0
$t^C_{2,2,2}$	440.01120358475225
$b^{H,\mathrm{in}}_{1,1,2}$	151.14362702928105
$b^{H,\mathrm{in}}_{1,2,1}$	4935.363246063736
$b^{H,\mathrm{in}}_{2,1,1}$	11800.0
$b^{H,\mathrm{in}}_{2,2,1}$	0.0
$b^{H,\mathrm{in}}_{2,2,2}$	5259.034878499343
$b^{H,\mathrm{out}}_{1,1,2}$	151.14362702928105
$b^{H,\mathrm{out}}_{1,2,1}$	4935.363246063736
$b^{H,\mathrm{out}}_{2,1,1}$	11800.0
$b^{H,\mathrm{out}}_{2,2,1}$	0.0

Variable	Value
$b^{H,\mathrm{out}}_{2,2,2}$	5259.034878499343
$b^{C,\mathrm{in}}_{1,1,2}$	1846.0835704276947
$b^{C,\mathrm{in}}_{1,2,1}$	6500.0
$b^{C,\mathrm{in}}_{2,1,1}$	2030.124334967239
$b^{C,\mathrm{in}}_{2,2,1}$	0.0
$b^{C,\mathrm{in}}_{2,2,2}$	0.0
$b^{C,\mathrm{out}}_{1,1,2}$	1846.0835704276951
$b^{C,\mathrm{out}}_{1,2,1}$	6500.0
$b^{C,\mathrm{out}}_{2,1,1}$	2030.124334967239
$b^{C,\mathrm{out}}_{2,2,1}$	0.0
$b^{C,\mathrm{out}}_{2,2,2}$	0.0